

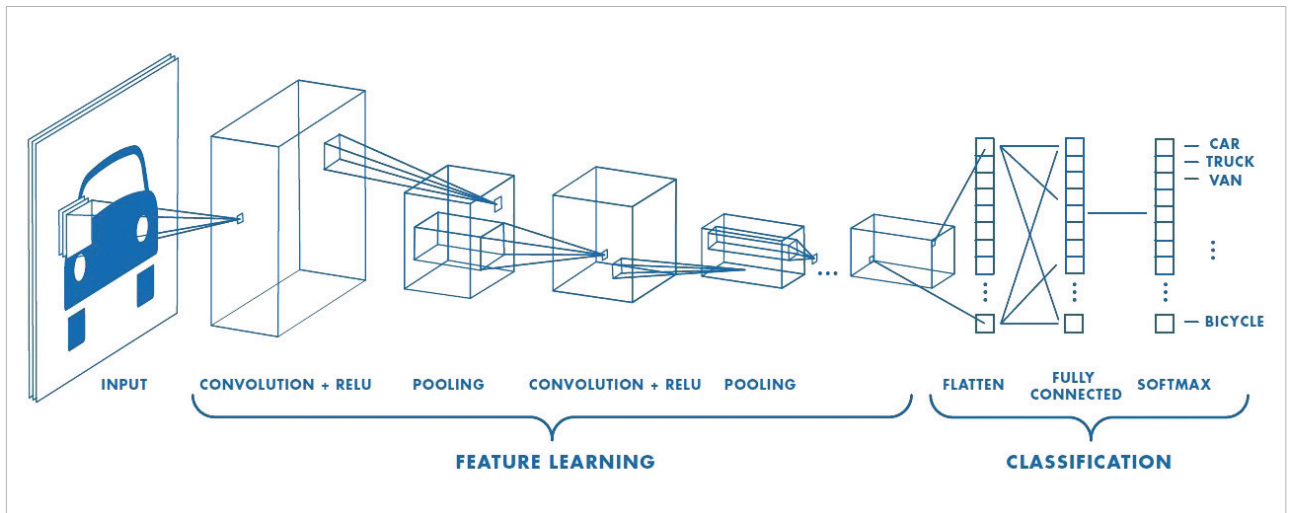


Figure 1: Basic CNN architecture

a composition. In 2009, when deep learning networks were becoming more interesting for research, Lee and Andrew Ng started using deep convoluted neural networks (CNN) for music genre classification. This formed a basis for advanced models that used high-level (semantic) concepts from music spectrograms. Most recently, Wavenet models and generative adversarial networks (GAN) have been used for generating music. Waveform-based models outperformed spectrogram-based ones, provided that enough training material was available.

LSTM, CNN and GAN

We shall now discuss the various ANN models that are used to create music and art.

LSTM networks are a variant of recurrent neural networks (RNN) that are capable of learning long term dependencies. The key to LSTMs is the cell state, which stores information. The ability to add to or remove information from the cell state is regulated by gates that are a sigmoid neural network layer coupled with a pointwise multiplication operation. Since music sequences are time series with long term dependencies, it is appropriate to use LSTMs.

CNN or convoluted neural networks are designed to process input images. Their architecture is composed of two main blocks. The first block functions as a feature extractor, which matches templates by applying convolution filtering operations. It returns feature maps that are normalised and/or resized. The second block constitutes the classification layer. The network constitutes an input layer, hidden layers and output layer. In CNNs, the hidden layers perform convolutions. This typically includes a layer that does multiplication or dot product with the *ReLU* activation function, followed by other convolution layers like the pooling layers and normalisation layers. A typical CNN architecture is shown in Figure 1.

GAN or generative adversarial networks have a generator,

discriminator and loss function. The role of the generator is to maximise the likelihood that the discriminator misclassifies its output as real. The discriminator's role is to optimise towards 50 per cent where it cannot identify between real and generated images. The generator starts training alongside the discriminator. The latter trains a few epochs prior to starting the adversarial training, as it will be required to actually classify images. The loss function provides the stopping criteria for the generator and discriminator training processes. Figures 2-4 are pictorial block diagrams of the GAN architecture, the GAN generator and the GAN discriminator. GANs are better than LSTM in producing music because they are able to capture the large structural patterns in the latter.

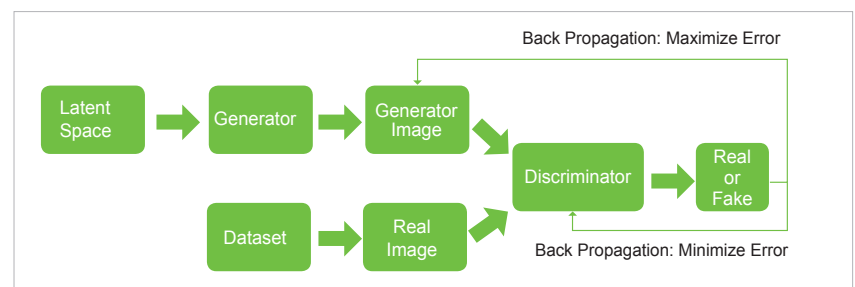


Figure 2: Basic GAN architecture